AUSTRALIAN MATHS TRUST

AUSTRALIAN MATHEMATICS COMPETITION

Chinese Version • Upper Primary

Australian school years 5, 6 & 7*

*Some states only

2019

NAME: _____

TIME ALLOWED: 60 minutes

INSTRUCTIONS AND INFORMATION

General

1 Do not open the booklet until told to do so by your teacher.

2 You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones are not permitted.

3 Diagrams are NOT drawn to scale. They are intended only as aids.

4 There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.

5 This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.

6 Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.

7 When your teacher gives the signal, begin working on the problems.

The answer sheet

1 Use only lead pencil.

2 Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.

3 Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder: You may sit this competition once, in one division only, or risk no score.

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1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 算式 $201 - 9 =$

$201 - 9 =$

(A) 111 (B) 182 (C) 188 (D) 192 (E) 198

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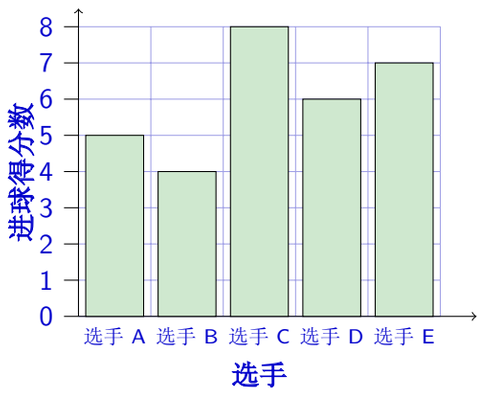
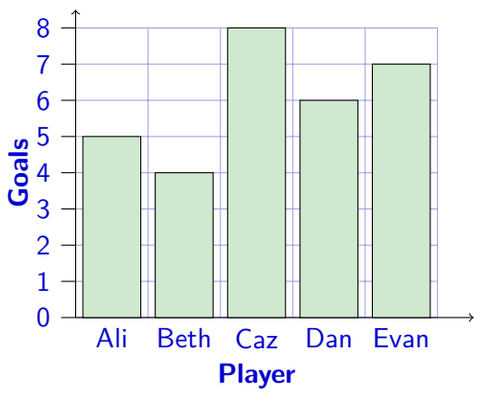
2. 某支球队有 5 位选手。
右表显示每位选手在某场球赛的进球得分数。
请问哪一个选项内选手的进球得分数第二高？

(A) 选手 A (B) 选手 B (C) 选手 C (D) 选手 D (E) 选手 E



A *Runnyball* team has 5 players.
This graph shows the number of goals each player scored in a tournament.
Who scored the second-highest number of goals?

(A) Ali (B) Beth (C) Caz (D) Dan (E) Evan



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3. 请问将陆佰贰拾万参仟零陆可以写成下列哪一个选项内的数？

Six million two hundred and three thousand and six would be written as

(A) 62036 (B) 6 230 006 (C) 6 203 006 (D) 6 203 600 (E) 6 200 306

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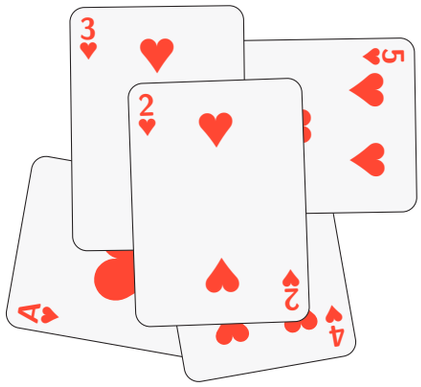
4. 这些牌一次一张地掉落在桌上。
请问它们掉落的顺序为哪一个选项内的顺序？

These cards were dropped on the table, one at a time.
In which order were they dropped?

(A) 4♥ A♥ 5♥ 3♥ 2♥ (B) A♥ 4♥ 5♥ 3♥ 2♥
(C) 2♥ 4♥ A♥ 3♥ 5♥ (D) A♥ 2♥ 3♥ 4♥ 5♥
(E) 2♥ 3♥ 4♥ 5♥ A♥



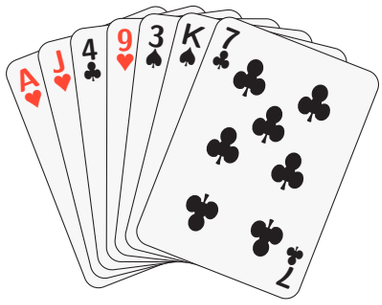
5. 小菲目前位于第 1 竖街与第 1 横路的交口。她的学校位于第 4 竖街与第 3 横路的交口。
请问她必须依照下列哪一个选项行走才能抵达学校？

(A) 向东走 4 段，再向北走 3 段
(B) 向西走 3 段，再向北走 4 段
(C) 向西走 4 段，再向北走 2 段
(D) 向东走 3 段，再向北走 2 段
(E) 向北走 2 段，再向南走 2 段

Sophia is at the corner of 1st Street and 1st Avenue. Her school is at the corner of 4th Street and 3rd Avenue.
To get there, she walks

(A) 4 blocks east, 3 blocks north
(B) 3 blocks west, 4 blocks north
(C) 4 blocks west, 2 blocks north
(D) 3 blocks east, 2 blocks north
(E) 2 blocks north, 2 blocks south

6. 小杰玩牌，这是他手上的牌。
小安从小杰手上的牌中随意抽出一张。
请问下列哪一个选项内的牌被小安抽到的机会最大？



(A) 一张红心 (♥)
(B) 一张方块 (♦)
(C) 一张黑桃 (♠)
(D) 一张有图片的牌 (J、Q 或 K)
(E) 一张偶数点的牌

Jake is playing a card game, and these are his cards.
Elena chooses one card from Jake at random.
Which of the following is Elena most likely to choose?

(A) a heart (♥) (B) a diamond (♦) (C) a spade (♠)
(D) a picture card (J, Q or K) (E) an even-numbered card

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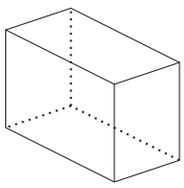
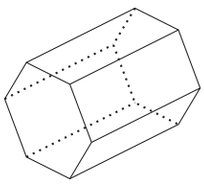
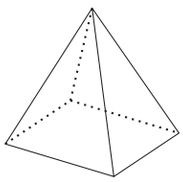
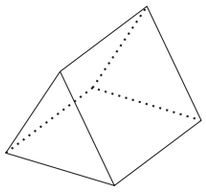
7. 请问下列哪一个选项内的 3D 立体图形有 5 个面且有 9 条边？

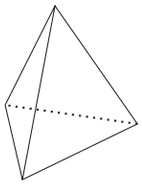
Which 3D shape below has 5 faces and 9 edges?

(A) 

(B) 

(C) 

(D) 

(E) 

8. 我们正驾车从 E 市到 R 市，刚离开时我们看到右侧之标示牌。
我们希望在前往 R 市接近一半路途的地方停下来午餐并休息。
请问在哪一个城市停下来最合适？

(A) G 市　(B) N 市　(C) T 市　(D) B 市　(E) W 市

We're driving from Elizabeth to Renmark, and as we leave we see this sign.
We want to stop at a town for lunch and a break, approximately halfway to Renmark.
Which town is the best place to stop?

(A) Gawler　(B) Nuriootpa　(C) Truro　(D) Blanchetown　(E) Waikerie

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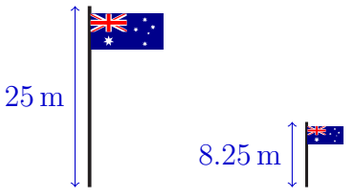
9. 请问这两根旗杆高度之差为多少 m ？

What is the difference between the heights of the two flagpoles, in metres?

(A) 16.25　(B) 16.75　(C) 17.25　(D) 17.75　(E) 33.25



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10. 在这个刻度尺上的大部份标示的数都不见了。

请问点 P 位置上所标示的数应该是什么？

Most of the numbers on this scale are missing.
Which number should be at position P?

(A) 18　(B) 33　(C) 34　(D) 36　(E) 42

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 在一个游戏中，投掷两枚标记为 0 到 9 的十面体骰子，然后将两个在最顶面的数相加。
例如图中所的两个骰子为 $0+9=9$。
请问我们总共可以得到多少种不同的结果？

In a game, two ten-sided dice each marked 0 to 9 are rolled and the two uppermost numbers are added.
For example, with the dice as shown, $0+9=9$.
How many different results can be obtained?

(A) 17 (B) 18 (C) 19 (D) 20 (E) 21

12. 在这个 3×3 的方格表的每一行每一列都必须包括数1、2、3各一个。
请问 $N+M$ 之值是什么？

1		
	2	N
		M

Every row and every column of this 3×3 square must contain each of the numbers 1, 2 and 3.
What is the value of $N+M$?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

13. 爱达·勒芙蕾丝 (Ada Lovelace) 与查尔斯·巴贝奇 (Charles Babbage) 是研究早期机械式计算机的先驱。他们的生日相距 24 年。

爱达·勒芙蕾丝 Ada Lovelace
1815 年 12 月–1852 年 11 月
December 1815–November 1852

查尔斯·巴贝奇 Charles Babbage
1791 年 12 月–1871 年 10 月
December 1791–October 1871

请问查尔斯·巴贝奇比爱达·勒芙蕾丝多活几年 (取至最接近的年)？

Ada Lovelace and Charles Babbage were pioneering researchers into early mechanical computers. They were born 24 years apart.

To the nearest year, how much longer did Charles Babbage live than Ada Lovelace?

(A) 29 (B) 32 (C) 35 (D) 37 (E) 43

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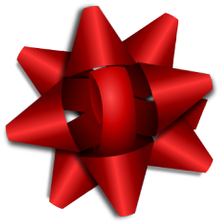
14. 您有 12 m 的彩带，每个装饰品需要使用 $\frac{2}{5}$ m 的彩带。
请问您总共可以做多少个装饰品？



You have 12 metres of ribbon. Each decoration needs $\frac{2}{5}$ of a metre of ribbon.
How many decorations can you make?

(A) 6 (B) 7 (C) 10 (D) 24 (E) 30

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15. 小安与小柏两兄弟清扫他家后院的落叶。
小柏可以用 60 分钟耙清后院，小安可以将吸尘器开成吹叶模式而在 30 分钟完成。
如果两人一起合作，请问需要多少分钟可以完成工作？

Andrew and Bernadette are clearing leaves from their backyard.
Bernadette can rake the backyard in 60 minutes, while Andrew can do it in 30 minutes with the vacuum setting on the leaf blower.
If they work together, how many minutes will it take?

(A) 10 (B) 20 (C) 24 (D) 30 (E) 45

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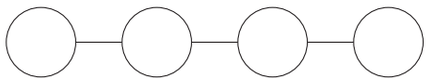
16. 每片地毯的尺寸为 50 cm × 50 cm。要铺满一间 6 m 长、4 m 宽的房间之地板，请问共需要多少片这样的地毯？

A carpet tile measures 50 cm by 50 cm. How many of these tiles would be needed to cover the floor of a room 6 m long and 4 m wide?

(A) 24 (B) 20 (C) 40 (D) 48 (E) 96

---

17. 将 1 到 4 的数分别不重复地填入这四个圆内，使得任两个连续的数都互不相邻。请问您总共有多少种不同的填法？



In how many different ways can you place the numbers 1 to 4 in these four circles so that no two consecutive numbers are side by side?

(A) 2 (B) 4 (C) 6 (D) 8 (E) 12

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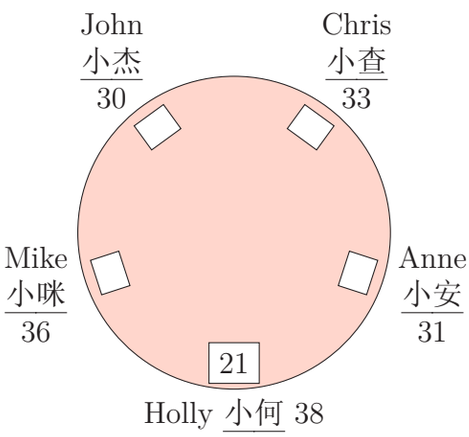
18. 小杰、小查、小安、小何、小咪围坐在一个圆桌前，每个人面前都有一张写有一个数的卡片。

每个人都可以看到他们两位邻居卡片上的数，每个人都喊出两位邻居卡片上的数之和。

已知小杰喊 30、小查喊 33、小安喊 31、小何喊 38、小咪喊 36。

若小何面前卡片的数是 21，请问小安面前卡片上的数是什么？



John, Chris, Anne, Holly and Mike are seated around a round table, each with a card with a number on it in front of them.

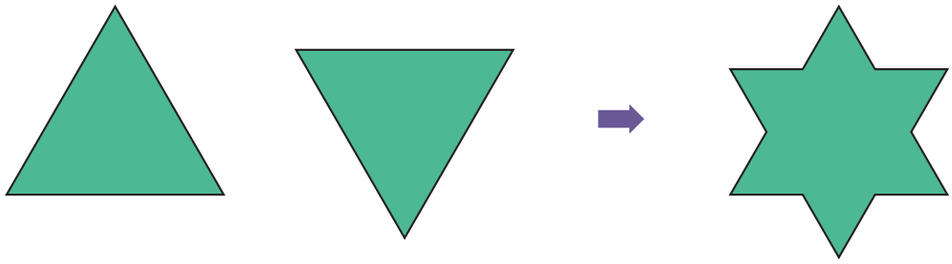
Each person can see the numbers in front of their two neighbours. Each person calls out the sum of the two numbers in front of their neighbours.

John says 30, Chris says 33, Anne says 31, Holly says 38 and Mike says 36.

Holly has the number 21 in front of her. What number does Anne have in front of her?

(A) 9 (B) 13 (C) 15 (D) 18 (E) 19

19. 小安有二个全等的正三角形，每个的面积都是 $9\,\text{cm}^2$。她将其中一个三角形放在另一个的上方构成一个星状形，如下图所示。请问这个星状形的面积为多少 cm^2？



Annabel has 2 identical equilateral triangles. Each has an area of $9\,\text{cm}^2$. She places one triangle on top of the other as shown to form a star, as shown. What is the area of the star in square centimetres?

(A) 10 (B) 12 (C) 14 (D) 16 (E) 18

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20. 小萝搭火车旅行。在行程中她睡了 $\frac{3}{4}$ 小时，清醒的时刻占全程的 $\frac{3}{4}$。请问她的这个行程总共费时多久？

(A) 1 小时 (B) 2 小时 (C) $2\frac{1}{2}$ 小时 (D) 3 小时 (E) 4 小时

Lola went on a train trip.
During her journey she slept for $\frac{3}{4}$ of an hour and stayed awake for $\frac{3}{4}$ of the journey.
How long did the trip take?

(A) 1 hour (B) 2 hours (C) $2\frac{1}{2}$ hours (D) 3 hours (E) 4 hours

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21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 我的姐姐与我玩一个游戏，她任选两个正整数由我来猜。当我告诉她一个数，她把我这个数乘以她的第一个数，然后加上第二个数。
当我说 15，她回答 50。当我说 2，她回答 11。
如果我说 6，她应该回答什么？

My sister and I are playing a game where she picks two counting numbers and I have to guess them. When I tell her a number, she multiplies my number by her first number and then adds her second number.
When I say 15, she says 50. When I say 2, she says 11.
If I say 6, what should she say?

(A) 23 (B) 27 (C) 35 (D) 41 (E) 61

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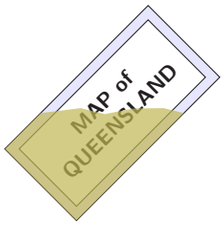
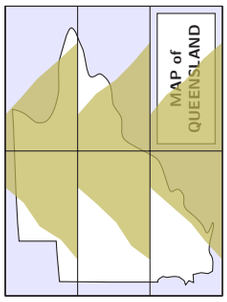
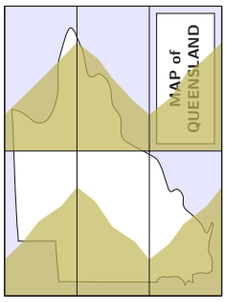
22. 当 2018 年洪水淹入小哈的家里时，他发现这张折叠的地图以一个角度浸入洪水中。
他将它展开并把它晾干，但地图仍然沾染了泥巴。
请问现在这张地图看起来的样子是下列哪一个选项？



Once the muddy water from the 2018 Ingham floods had drained from Harry's house, he found this folded map that had been standing in the floodwater at an angle.
He unfolded it and laid it out to dry, but it was still mud-stained.
What could it look like now?

(A)


(B)


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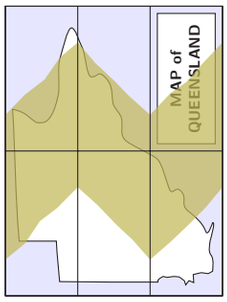

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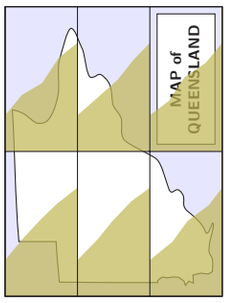

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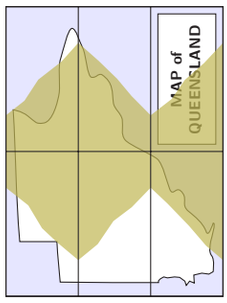
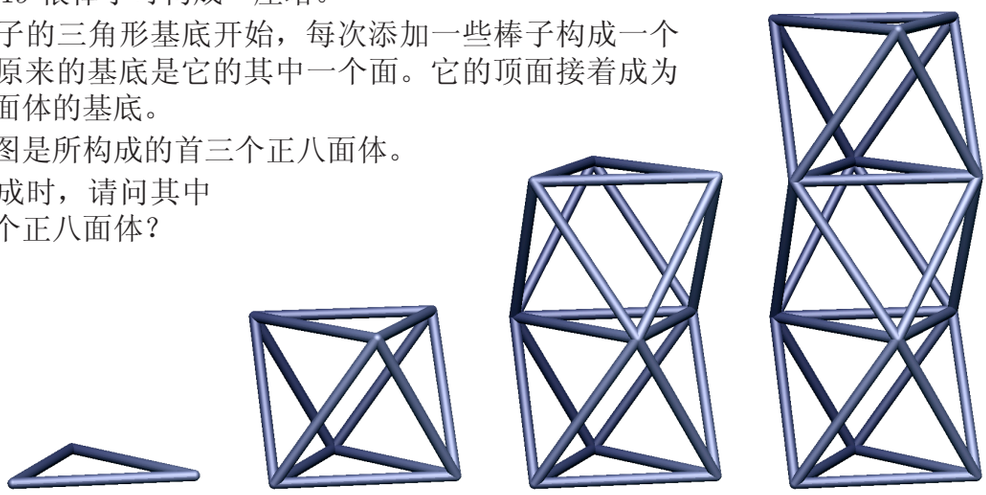


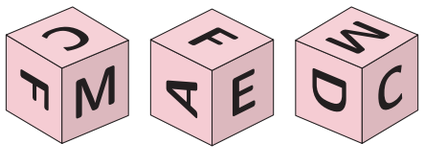
23. 恰好使用 2019 根棒子可构成一座塔。
从用三根棒子的三角形基底开始，每次添加一些棒子构成一个正八面体，原来的基底是它的其中一个面。它的顶面接着成为下一个正八面体的基底。
以下所示的图是所构成的首三个正八面体。
当这座塔完成时，请问其中总共有多少个正八面体？



A tower is built from exactly 2019 equal rods.
Starting with 3 rods as a triangular base, more rods are added to form a regular octahedron with this base as one of its faces. The top face is then the base of the next octahedron.
The diagram shows the construction of the first three octahedra.
How many octahedra are in the tower when it is finished?

(A) 2016 (B) 1008 (C) 336 (D) 224 (E) 168

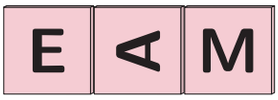
24. 这三个正立方体上的标记都完全相同，其中 6 个英文字母 A、M、C、D、E、F 恰在它们的 6 个面上。

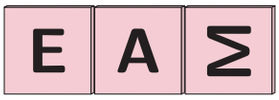


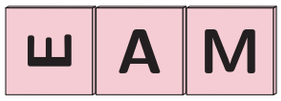
现在将这些正立方体排成一列，使得从正前方看过去它们的样子如下：

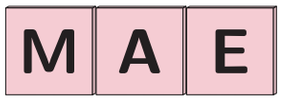
当我们从这些正立方体的正后方看过去时，请问我们将看到什么？

(A)

(B) 

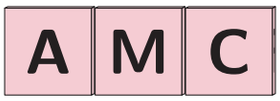
(C) 

(D) 

(E) 

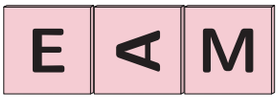
These three cubes are labelled in exactly the same way, with the 6 letters A, M, C, D, E and F on their 6 faces:

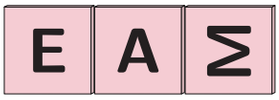
The cubes are now placed in a row so that the front looks like this:

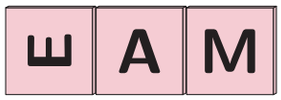


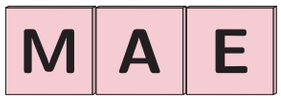
When we look at the cubes from the opposite side, we will see

(A)

(B) 

(C) 

(D) 

(E) 

25. 在游泳嘉年华的前四名将得到分数，如右表所示。
阿里参加自由式、仰式、蛙式，共得到 11 分。
请问阿里有多少种不同的方式可得到这样的分数？

	第一名	7 分
	第二名	5 分
	第三名	3 分
	第四名	1 分

Points are awarded for the first four places at the swimming carnival as shown in the table.
Aalia competed in the freestyle, backstroke and breaststroke races and scored a total of 11 points.
In how many different ways could she have done this?

(A) 6 (B) 9 (C) 12 (D) 18 (E) 24

	1st place	7 points
	2nd place	5 points
	3rd place	3 points
	4th place	1 point

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 小伦与小乃构造一列数。他们从 0 开始，小伦加上 20，接着小乃减 1，接着小伦加 19，接着小乃减 2。

$$0, 20, 19, 38, 36, \ldots$$

依此方式继续下去，直到小伦加 1 且小乃减 20 为止，最后将又回到 0。请问在此数列中最大的数是什么？

Warren and Naida make a list of numbers. They start with 0, then Warren adds 20, then Naida subtracts 1, then Warren adds 19, then Naida subtracts 2.

$$0, 20, 19, 38, 36, \ldots$$

This pattern continues until Warren adds 1 and Naida subtracts 20, which ends up back at 0. What is the largest number in their list?

27. 使用数码 1、5、6、8 与一个小数点可以造出不同的数。若每个数都恰使用一次且小数点必须介于两个数码之间，请问共可能造出多少个不同的数？

Different numbers can be made using the digits 1, 5, 6, 8 and a decimal point. How many possibilities are there if each digit must be used exactly once and the decimal point must lie between two digits?

---

28. 约翰比他的妻子玛丽大一岁。他们有三位小孩，小孩之间的年龄间隔为二岁。
约翰与玛丽年龄的乘积小于 2019，三位小孩年龄的乘积也小于 2019。
明年，这些乘积都将大于 2020。
请问这五个人今年的年龄总和为多少？

John is one year older than his wife Mary. They have three children, whose ages are two years apart.
The product of John and Mary's ages is less than 2019. The product of the three children's ages is also less than 2019.
Next year both these products will be greater than 2020.
This year, what is the sum of all five ages?

---

29. 有一个形如 $20AMC19$ 的七位数，它的所有的数码都互不相同。若这个七位数可以被 9 整除，请问 $A+M+C$ 之值是什么？

A seven-digit number is in the form $20AMC19$, with all digits different. It is divisible by 9.
What is the value of $A+M+C$?

---

30. 我有五位小孩，现打算将他们排成一列照相。但是小虎拒绝排在小莉与小理之间的任何位置。请问总共有多少种安排这些小孩排成一列且使小虎也满意的方法？

I have five children and want to place them in a line for a photograph. However, Hugh refuses to stand anywhere in between Louise and Richard. How many ways are there to place the children in a line and still keep Hugh happy?

草稿纸
Scratch Paper

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